

Details for Query 3

Submitted Time: 2026/09/16 16:29:06

Duration: 34 s

Succeeded Jobs: [5](#) [6](#) [7](#)

☒ Show the Stage ID and Task ID that corresponds to the max metric

☐ Show all the metrics in the plan

Native Engine Stages

Spark Stages

Scan parquet +details	
Metric	Value
number of files read	163
scan time	total (min, med, max (stageld: taskId)) 41.7 s (2.0 s, 4.1 s, 4.3 s (stage 5.0: task 9))
metadata time	0 ms
size of files read	862.7 MiB
number of output rows	39,790,092

Scan parquet +details	
Metric	Value
number of files read	1
scan time	133 ms
metadata time	0 ms
size of files read	149.9 KiB
number of output rows	5,000

WholeStageCodegen (1)

duration: total (min, med, max (stageld: taskId))
2.2 m (9.8 s, 11.8 s, 12.7 s (stage 5.0: task 9))
Stage: [5.0](#)

ColumnnarToRow +details

Metric	Value
number of output rows	39,790,092
number of input batches	9,796

Filter +details

Metric	Value
number of output rows	39,790,092

WholeStageCodegen (2)

duration: 267 ms

ColumnnarToRow +details

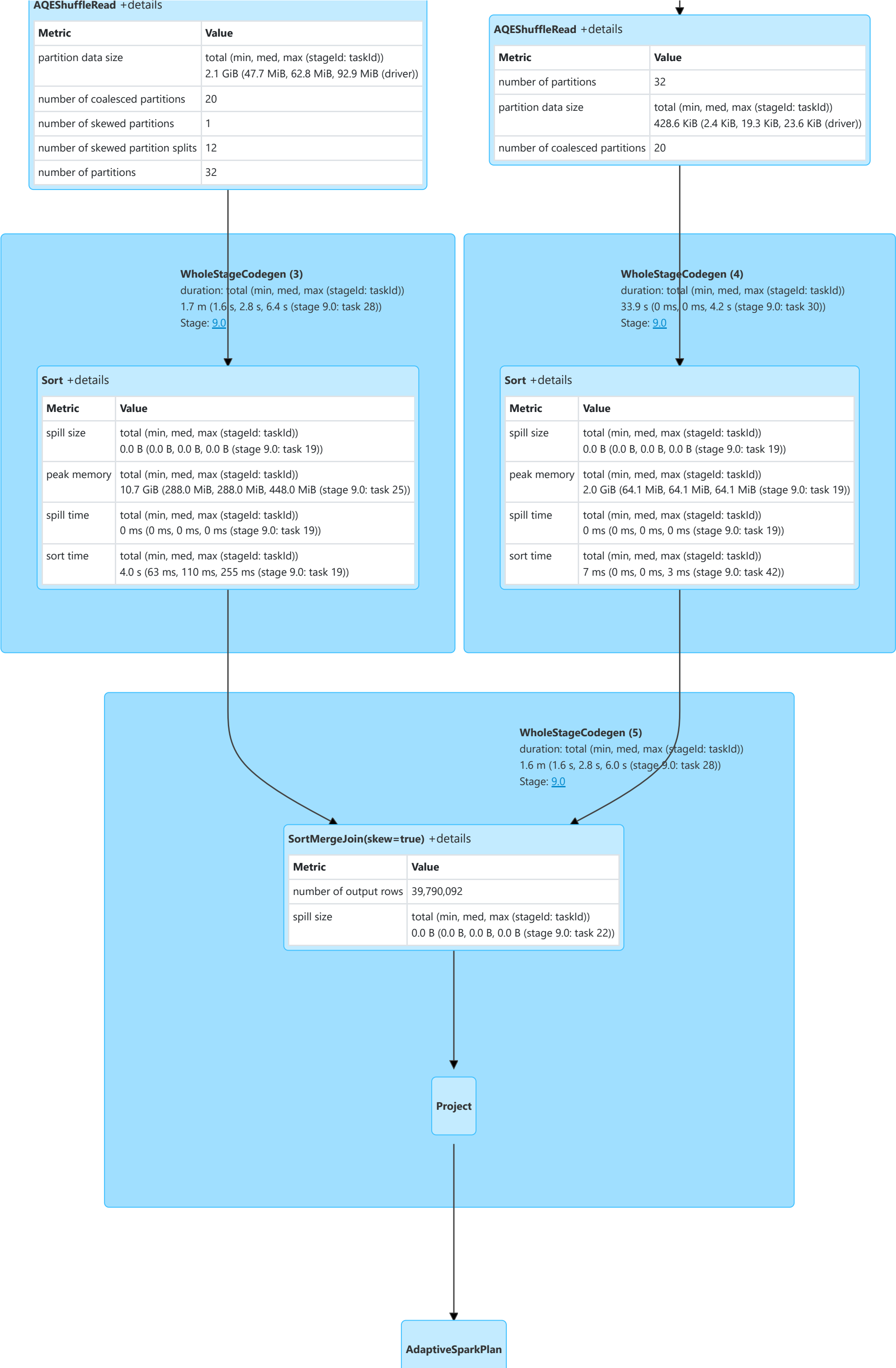
Metric	Value
number of output rows	5,000
number of input batches	2

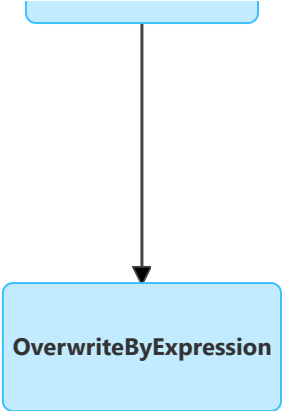
Filter +details

Metric	Value
number of output rows	5,000

Exchange +details	
Metric	Value
shuffle records written	39,790,092
shuffle write time	total (min, med, max (stageld: taskId)) 4.8 s (275 ms, 374 ms, 542 ms (stage 5.0: task 7))
records read	39,790,092
local bytes read	total (min, med, max (stageld: taskId)) 2.0 GiB (45.5 MiB, 59.8 MiB, 88.5 MiB (stage 9.0: task 28))
fetch wait time	total (min, med, max (stageld: taskId)) 0 ms (0 ms, 0 ms, 0 ms (stage 9.0: task 20))
local blocks read	252
data size	total (min, med, max (stageld: taskId)) 8.5 GiB (547.7 MiB, 744.7 MiB, 800.7 MiB (stage 5.0: task 6))
number of partitions	200
shuffle bytes written	total (min, med, max (stageld: taskId)) 2.0 GiB (130.4 MiB, 177.9 MiB, 191.4 MiB (stage 5.0: task 6))

Exchange +details	
Metric	Value
shuffle records written	5,000
shuffle write time	31 ms
records read	5,308
local bytes read	total (min, med, max (stageld: taskId)) 408.6 KiB (2.3 KiB, 18.5 KiB, 22.5 KiB (stage 9.0: task 43))
fetch wait time	total (min, med, max (stageld: taskId)) 0 ms (0 ms, 0 ms, 0 ms (stage 9.0: task 19))
local blocks read	32
data size	712.3 KiB
number of partitions	200
shuffle bytes written	383.8 KiB





▼ Details

```
== Physical Plan ==
OverwriteByExpression (26)
+- AdaptiveSparkPlan (25)
  +- == Final Plan ==
    * Project (16)
      +- * SortMergeJoin(skew=true) Inner (15)
        :- * Sort (7)
          : +- AQEShuffleRead (6)
            :   +- ShuffleQueryStage (5), Statistics(sizeInBytes=8.5 GiB, rowCount=3.98E+7)
              :     +- Exchange (4)
                :       +- * Filter (3)
                  :         +- * ColumnarToRow (2)
                    :           +- Scan parquet (1)
      +- * Sort (14)
        +- AQEShuffleRead (13)
          +- ShuffleQueryStage (12), Statistics(sizeInBytes=712.3 KiB, rowCount=5.00E+3)
            +- Exchange (11)
              +- * Filter (10)
                +- * ColumnarToRow (9)
                  +- Scan parquet (8)

  +- == Initial Plan ==
    Project (24)
      +- SortMergeJoin Inner (23)
        :- Sort (19)
          : +- Exchange (18)
            :   +- Filter (17)
              :     +- Scan parquet (1)
      +- Sort (22)
        +- Exchange (21)
          +- Filter (20)
            +- Scan parquet (8)
```

(1) Scan parquet
Output [11]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10]
Batched: true
Location: InMemoryFileIndex [abfss://testcontainer@storageinswa01us2dev.dfs.core.windows.net/pyspark_optimization/data/data_skew/transactions.parquet]
PushedFilters: [IsNotNull(cust_id)]
ReadSchema: struct<cust_id:string,start_date:string,end_date:string,txn_id:string,date:string,year:string,month:string,day:string,expense_type:string,amt:string,cit y:string>

(2) ColumnarToRow [codegen id : 1]
Input [11]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10]

(3) Filter [codegen id : 1]
Input [11]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10]
Condition : isnotnull(cust_id#0)

(4) Exchange
Input [11]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10]
Arguments: hashpartitioning(cust_id#0, 200), ENSURE_REQUIREMENTS, [plan_id=117]

(5) ShuffleQueryStage
Output [11]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10]
Arguments: 0

(6) AQEShuffleRead
Input [11]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10]
Arguments: coalesced and skewed

(7) Sort [codegen id : 3]
Input [11]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10]
Arguments: [cust_id#0 ASC NULLS FIRST], false, 0

(8) Scan parquet
Output [7]: [cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Batched: true
Location: InMemoryFileIndex [abfss://testcontainer@storageinswa01us2dev.dfs.core.windows.net/pyspark_optimization/data/data_skew/customers.parquet]
PushedFilters: [IsNotNull(cust_id)]
ReadSchema: struct<cust_id:string,name:string,age:string,gender:string,birthday:string,zip:string,city:string>

(9) ColumnarToRow [codegen id : 2]
Input [7]: [cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]

(10) Filter [codegen id : 2]
Input [7]: [cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Condition : isnotnull(cust_id#22)

(11) Exchange
Input [7]: [cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Arguments: hashpartitioning(cust_id#22, 200), ENSURE_REQUIREMENTS, [plan_id=139]

(12) ShuffleQueryStage
Output [7]: [cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Arguments: 1

(13) AQEShuffleRead
Input [7]: [cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Arguments: coalesced

(14) Sort [codegen id : 4]
Input [7]: [cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Arguments: [cust_id#22 ASC NULLS FIRST], false, 0

(15) SortMergeJoin(skew=true) [codegen id : 5]
Left keys [1]: [cust_id#0]
Right keys [1]: [cust_id#22]
Join type: Inner
Join condition: None

(16) Project [codegen id : 5]
Output [17]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Input [18]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10, cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]

(17) Filter
Input [11]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10]
Condition : isnotnull(cust_id#0)

(18) Exchange
Input [11]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10]
Arguments: hashpartitioning(cust_id#0, 200), ENSURE_REQUIREMENTS, [plan_id=98]

(19) Sort
Input [11]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10]
Arguments: [cust_id#0 ASC NULLS FIRST], false, 0

(20) Filter
Input [7]: [cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Condition : isnotnull(cust_id#22)

(21) Exchange
Input [7]: [cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Arguments: hashpartitioning(cust_id#22, 200), ENSURE_REQUIREMENTS, [plan_id=99]

(22) Sort
Input [7]: [cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Arguments: [cust_id#22 ASC NULLS FIRST], false, 0

(23) SortMergeJoin
Left keys [1]: [cust_id#0]
Right keys [1]: [cust_id#22]
Join type: Inner
Join condition: None

(24) Project
Output [17]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Input [18]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10, cust_id#22, name#23, age#24, gender#25, birthday#26, zip#27, city#28]

(25) AdaptiveSparkPlan
Output [17]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Arguments: isFinalPlan=true

(26) OverwriteByExpression
Input [17]: [cust_id#0, start_date#1, end_date#2, txn_id#3, date#4, year#5, month#6, day#7, expense_type#8, amt#9, city#10, name#23, age#24, gender#25, birthday#26, zip#27, city#28]
Arguments: org.apache.spark.sql.execution.datasources.v2.DataSourceV2Strategy\$\$Lambda\$5889/0x000074779373d6a8@75b052b0, org.apache.spark.sql.execution.datasources.noop.NoopWrite\$@5f3d2c7c

▼ SQL / DataFrame Properties

Name	Value
spark.app.name	SynapseEnvPy
spark.sql.autoBroadcastJoinThreshold	-1
spark.sql.parquet.fieldId.read.enabled	true
spark.sql.parquet.fieldId.write.enabled	true

▼Stage Metrics Summary

Stage Name	Stage Id	Native Execution Engine	Metric	Value
AQEShuffleRead	stage 9.0	No	partition data size	total (min, med, max (stageld: taskld))2.1 GiB (47.7 MiB, 62.8 MiB, 92.9 MiB (driver))
			number of coalesced partitions	20
			number of skewed partitions	1
			number of skewed partition splits	12
			number of partitions	32
			number of partitions	32
			partition data size	total (min, med, max (stageld: taskld))428.6 KiB (2.4 KiB, 19.3 KiB, 23.6 KiB (driver))
			number of coalesced partitions	20
ColumnarToRow	stage 5.0	No	number of output rows	39,790,092
			number of input batches	9,796
ColumnarToRow	stage 9.0	No	number of output rows	5,000
			number of input batches	2
Exchange	stage 5.0	No	shuffle records written	39,790,092
			shuffle write time	total (min, med, max (stageld: taskld))4.8 s (275 ms, 374 ms, 542 ms (stage 5.0: task 7))
			records read	39,790,092
			local bytes read	total (min, med, max (stageld: taskld))2.0 GiB (45.5 MiB, 59.8 MiB, 88.5 MiB (stage 9.0: task 28))
			fetch wait time	total (min, med, max (stageld: taskld))0 ms (0 ms, 0 ms, 0 ms (stage 9.0: task 20))
			local blocks read	252
			data size	total (min, med, max (stageld: taskld))8.5 GiB (547.7 MiB, 744.7 MiB, 800.7 MiB (stage 5.0: task 6))
			number of partitions	200
			shuffle bytes written	total (min, med, max (stageld: taskld))2.0 GiB (130.4 MiB, 177.9 MiB, 191.4 MiB (stage 5.0: task 6))
Exchange	stage 9.0	No	shuffle records written	5,000
			shuffle write time	31 ms
			records read	5,308
			local bytes read	total (min, med, max (stageld: taskld))408.6 KiB (2.3 KiB, 18.5 KiB, 22.5 KiB (stage 9.0: task 43))
			fetch wait time	total (min, med, max (stageld: taskld))0 ms (0 ms, 0 ms, 0 ms (stage 9.0: task 19))
			local blocks read	32
			data size	712.3 KiB
			number of partitions	200
			shuffle bytes written	383.8 KiB
Filter	stage 5.0	No	number of output rows	39,790,092
Filter	stage 9.0	No	number of output rows	5,000

Stage Name	Stage Id	Native Execution Engine	Metric	Value
Scan parquet	stage 5.0	No	number of files read	163
			scan time	total (min, med, max (stageld: taskld))41.7 s (2.0 s, 4.1 s, 4.3 s (stage 5.0: task 9))
			metadata time	0 ms
			size of files read	862.7 MiB
			number of output rows	39,790,092
Scan parquet	stage 9.0	No	number of files read	1
			scan time	133 ms
			metadata time	0 ms
			size of files read	149.9 KiB
			number of output rows	5,000
Sort	stage 9.0	No	spill size	total (min, med, max (stageld: taskld))0.0 B (0.0 B, 0.0 B, 0.0 B (stage 9.0: task 19))
			peak memory	total (min, med, max (stageld: taskld))10.7 GiB (288.0 MiB, 288.0 MiB, 448.0 MiB (stage 9.0: task 25))
			spill time	total (min, med, max (stageld: taskld))0 ms (0 ms, 0 ms, 0 ms (stage 9.0: task 19))
			sort time	total (min, med, max (stageld: taskld))4.0 s (63 ms, 110 ms, 255 ms (stage 9.0: task 19))
			spill size	total (min, med, max (stageld: taskld))0.0 B (0.0 B, 0.0 B, 0.0 B (stage 9.0: task 19))
			peak memory	total (min, med, max (stageld: taskld))2.0 GiB (64.1 MiB, 64.1 MiB, 64.1 MiB (stage 9.0: task 19))
			spill time	total (min, med, max (stageld: taskld))0 ms (0 ms, 0 ms, 0 ms (stage 9.0: task 19))
			sort time	total (min, med, max (stageld: taskld))7 ms (0 ms, 0 ms, 3 ms (stage 9.0: task 42))
SortMergeJoin(skew=true)	stage 9.0	No	number of output rows	39,790,092
			spill size	total (min, med, max (stageld: taskld))0.0 B (0.0 B, 0.0 B, 0.0 B (stage 9.0: task 22))